

STALKS OF AUTOMORPHIC VOGAN SHEAVES FOR THE STEINBERG PARAMETER OF GL_n

1. THE MAIN CONJECTURE

1.1. Vogan sheaf. We first fix notation to define the Vogan stack, and then construct the Vogan sheaf with weight.

Notation 1.1. We use the same notation as in David's talk.

- For $n \geq 2$, take $G = \mathrm{GL}_n$ over a finite extension $E/\mathbb{Q}_p$ with residue field $\mathbb{F}_q$. Take coefficient ring $\Lambda = \overline{\mathbb{Q}}_\ell$ with $\ell \neq p$ and fix $q^{1/2} \in \Lambda$.
- Let $T \subset B \subset G$ be the maximal split torus and the standard Borel in G . For $1 \leq i \leq n-1$, denote by $\alpha_i := e_i - e_{i+1} \in \mathbf{X}^*(\hat{T})$ the simple roots of GL_n .
- Let φ be the semisimple L-parameter such that $\varphi(I_F) = 1$ and $\varphi(\mathrm{Fr}) = \delta^{1/2}$, where Fr is the geometric q -Frobenius and

$$\delta^{1/2} := \mathrm{diag}(q^{(1-n)/2}, \dots, q^{(n-1)/2}) \in \hat{T}(\Lambda).$$

Note that the finite Weyl group $W = S_n$ acts on $\delta^{1/2}$ by permuting the diagonal elements, so we get the G -conjugacy class $[\varphi] \in (\hat{T} // W)(\Lambda)$.

With Notation 1.1, we construct the Vogan sheaf $\mathcal{L}_{\mathbf{k}}$ as follows. Recall from [Han23, §1.3] that the Vogan stack $V_{\hat{G}, \varphi}$ at φ parametrizes nilpotent elements $N \in \mathfrak{g}^{\mathrm{ad} \varphi(I_F)}$ such that $\mathrm{ad} \varphi(\mathrm{Fr}) \cdot N = q^{-1}N$ up to S_φ -conjugacy, where $S_\varphi := \mathrm{Cent}_{\hat{G}}(\varphi)$. For $G = \mathrm{GL}_n$, we have

$$\begin{array}{ccc} V_{\hat{G}, \varphi} = \mathbb{A}^{n-1} / \hat{T} & \xhookrightarrow{\iota} & \mathrm{Par}_{\hat{G}}^{\mathrm{unip}} \\ \downarrow & & \\ \mathbb{B}\hat{T} & & \end{array}$$

as a closed substack of the stack of unipotent L-parameters for $\hat{G}$. This diagram depends only on $[\varphi]$. Note that ι factors through $\mathrm{Par}_{\hat{G}}^{\mathrm{ur}} \hookrightarrow \mathrm{Par}_{\hat{G}}^{\mathrm{unip}}$ because of the condition $\varphi(I_F) = 1$.

We then fix $\mathbf{k} = (k_1, \dots, k_n) \in \mathbf{X}^*(\hat{T})$ and consider its associated weight character

$$\chi_{\mathbf{k}} : \mathrm{diag}(t_1, \dots, t_n) \mapsto t_1^{k_1} \cdots t_n^{k_n}$$

on $\hat{T} = \mathbb{G}_m^n$, valued in $\Lambda^\times$. Viewing this $\chi_{\mathbf{k}}$ as a coherent sheaf on $\mathbb{B}\hat{T}$, its pull-push along the diagram above yields a coherent sheaf

$$\mathcal{L}_{\mathbf{k}} := \iota_* \mathcal{O}(\mathbf{k}) \in \mathrm{Coh}(\mathrm{Par}_{\hat{G}}^{\mathrm{unip}}),$$

where $\mathcal{O}(\mathbf{k})$ on $\mathbb{A}^{n-1} / \hat{T}$ is the pullback of $\chi_{\mathbf{k}}$ from $\mathbb{B}\hat{T}$.

1.2. The Steinberg stalk. Assume the categorical local Langlands equivalence. Then $\mathcal{L}_{\mathbf{k}}$ corresponds to an automorphic sheaf $\mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}$ on Bun_G . As we are interested in the smooth irreducible representation corresponding to $\mathcal{L}_{\mathbf{k}}$, we aim to compute the stalk $i_1^* \mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}$, where $i_1 : \mathrm{Rep}(G(E), \Lambda) \rightarrow \mathrm{D}_{\mathrm{lis}}(\mathrm{Bun}_G, \Lambda)$. More generally, one can also consider $i_b^* \mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}$ for basic $b \in \mathrm{B}(G)_{\mathrm{bas}}$, with $i_b : \mathrm{Rep}(G_b(E), \Lambda) \rightarrow \mathrm{D}_{\mathrm{lis}}(\mathrm{Bun}_G^b, \Lambda)$. In the GL_n case, this is called the *Steinberg stalk* as it turns out to be some generalized Steinberg representation π_{I} defined as follows, up to a cohomological degree shift (see Conjecture 1.4).

Definition 1.2. Given a subset $I \subset \{1, \dots, n-1\}$ with elements $i_1 < \dots < i_k$, we obtain the standard parabolic subgroup P_I with Levi $\mathrm{GL}_{i_1} \times \mathrm{GL}_{i_2-i_1} \times \dots \times \mathrm{GL}_{i_k-i_{k-1}} \times \mathrm{GL}_{n-i_k}$. Define the *generalized Steinberg representation* π_I as the unique irreducible quotient of $\mathcal{C}(P_I(E) \backslash G(E), \Lambda)$.

Note that the π_1 's are exactly the irreducible representations whose semisimple L-parameter is φ (corresponding to $\delta^{1/2}$). It is a standard fact that the map $I \mapsto \pi_1$ assigned by Definition 1.2 above is actually a canonical bijection.

We point out that in Zhu's context [Zhu25], the construction of $\mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}$ from $\mathcal{L}_{\mathbf{k}}$ is unconditional at the tame level. In this note, we work for simplicity at the unipotent level, and then Zhu's proof yields an equivalence functor $\mathbb{L}^{\mathrm{unip}}$. To apply Zhu's result, we need to change Bun_G to Isoc_G , and replace D_{lis} with the sheaf theory on Isoc_G . Moreover, the stalk $i_{\mathbf{b}}^* \mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}$ must be written as $i_{\mathbf{b}}^! \mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}$ instead; see [Zhu25, 1.2.1, 1.2.3] for more background.

1.3. Inner forms. Pick any integer d and let $\mathbf{b}_{d/n} = \mathbf{b}_d$ be the basic isocrystal of slope d/n . Write $n' := n/(d, n)$ and $d' := d/(d, n)$ (where we declare $(0, n) = n$). Take $D_{d'/n'}$ as the unique central division algebra over E with Brauer invariant $d'/n' \in \mathbb{Q}/\mathbb{Z}$. Then $G_{\mathbf{b}_d} = \mathrm{GL}_{(d, n)}(D_{d'/n'})$ gives an inner form of G (recall that for $G = \mathrm{GL}_n$ over E , any inner form must be of the form $\mathrm{GL}_m(D)$ with $\deg D = n/m$). Note that $G_{\mathbf{b}_d}$ depends only on $d \bmod n$, because the image of Brauer invariant d'/n' in $\mathbb{Q}/\mathbb{Z}$ depends only on $d \bmod n$.

Fact 1.3. There is a canonical bijection between generalized Steinberg representations of $G_{\mathbf{b}_d}$ and subsets $I \subset \{1, \dots, (d, n) - 1\}$.

Due to this fact, we can generalize the previous construction $I \mapsto \pi_1$ from representations of $G(E)$ to representations of $G_{\mathbf{b}_d}(E)$.

1.4. The main conjecture. Fix integers d, n as before. For $\mathbf{k} = (k_1, \dots, k_n) \in \mathbf{X}^*(\hat{T})$ of degree d , i.e., $k_1 + \dots + k_n = d$, there are unique integers $m_i \in \mathbb{Z}$ such that

$$\mathbf{k} = \boldsymbol{\omega}_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i.$$

Here $\boldsymbol{\omega}_d \in \mathbf{X}^*(\hat{T})$ is called the *Steinberg weight vector* of $G_{\mathbf{b}_d}$, defined as

$$\boldsymbol{\omega}_d := ([d/n], [2d/n] - [d/n], \dots, d - [(n-1)d/n]).$$

Notice that $\boldsymbol{\omega}_d$ is an n -tuple of degree d . In particular, $\mathbf{k} - \boldsymbol{\omega}_d$ is of degree 0.

With the notations above, we state the main conjecture as follows.

Conjecture 1.4 (Hansen). *Fix n, d as before. Suppose $\mathbf{k}$ has degree d . For each $1 \leq i \leq n-1$, define $\delta_i \in \{0, 1\}$ by setting $\delta_i = 1$ if and only if $n' = n/(d, n)$ divides i . Then we expect*

$$i_{\mathbf{b}_d}^* \mathcal{L}_{\mathbf{k}}^{\mathrm{aut}} = \pi_{\mathbf{I}_{\mathbf{k}}} \left[\sum_{j \in \mathbf{J}_{\mathbf{k}}} (\delta_j - 2m_j) \right]$$

between derived complexes in $\mathrm{Rep}(G_{\mathbf{b}_d}(E), \Lambda)$ that are concentrated in degree 0. Here,

- $\mathbf{I}_{\mathbf{k}} := \{i \in \{1, \dots, (d, n) - 1\} \mid m_{n'i} \leq 0\}$;
- $\mathbf{J}_{\mathbf{k}} := \{i \in \{1, \dots, n-1\} \mid m_i > 0\}$;
- $\pi_{\mathbf{I}_{\mathbf{k}}}$ is the unique generalized Steinberg representation of $G_{\mathbf{b}_d}(E)$ corresponding to $\mathbf{I}_{\mathbf{k}}$.

In fact, the stalk $i_{\mathbf{b}_d}^* \mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}$ is identically zero unless $\mathbf{k}$ has degree d . To check this, note that the central character $\omega_{\mathbf{k}}$ of $\mathbf{k}$ on $Z(\hat{G}) = \mathbb{G}_m$ is given by $\omega_{\mathbf{k}}: z \mapsto z^{k_1 + \dots + k_n}$; on the other hand, along the Kottwitz map $\kappa_G: \mathrm{B}(G) \rightarrow \mathbf{X}_*(Z(\hat{G})^\Gamma)$, we always have $\kappa_G(\mathbf{b}_d) = d \bmod n$ for $G = \mathrm{GL}_n$.

2. PROOF OF CONJECTURE 1.4 FOR $\mathbf{b} = 1$

In this section, we prove Conjecture 1.4 for the special case $\mathbf{b} = 1$, achieved by taking $d = 0$. Note that when $d = 0$, we have the following ingredients in practice.

- $G_{\mathbf{b}_d} = G = \mathrm{GL}_n$.
- $(d, n) = (0, n) = n$, which forces $\mathbf{I}_{\mathbf{k}} = \{i \in \{1, \dots, n-1\} \mid m_i \leq 0\} = \{1, \dots, n-1\} \setminus \mathbf{J}_{\mathbf{k}}$ to hold, and $\delta_i = 1$ for all $1 \leq i \leq n-1$.
- $\mathbf{k} = (k_1, \dots, k_n)$ is of degree $k_1 + \dots + k_n = 0$ (with this degree condition, we attain $\mu_{\mathbf{k}} := k_1 e_1 + \dots + k_n e_n = k_1 \alpha_1 + (k_1 + k_2) \alpha_2 + \dots + (k_1 + \dots + k_{n-1}) \alpha_{n-1}$ by substituting $\alpha_j = e_j - e_{j+1}$, and thus $m_j = k_1 + \dots + k_j$).

Therefore, Conjecture 1.4 for $b = 1$ can be simplified into the following statement.

Theorem 2.1. *In $\text{Rep}(GL_n(E), \Lambda)$, for $\mathbf{k} = (k_1, \dots, k_n)$ of degree 0, we have*

$$i_1^* \mathcal{L}_{\mathbf{k}}^{\text{aut}} = \pi_{I_{\mathbf{k}}} \left[\sum_{j \notin I_{\mathbf{k}}} (1 - 2m_j) \right],$$

where $m_j = k_1 + \dots + k_j$.

The main strategy to prove Theorem 2.1 is first reducing to computation of some RHom -complex of derived sheaves on Steinberg stack by arguing with Jacquet module, and then constructing a free projective resolution to get the cohomology of this RHom -complex.

2.1. Jacquet module and coherent Springer sheaf. Recall that we have defined the semisimple L-parameter φ by $\delta^{1/2}$, whose corresponding smooth irreducible representations of $GL_n(E)$ are exactly π_I 's (where the monodromy of the Weil–Deligne parameter is parametrized by I). Thus, the stalk $i_1^* \mathcal{L}_{\mathbf{k}}^{\text{aut}}$ must be a Steinberg representation up to shift. On the other hand, each π_I can be characterized among all generalized Steinberg representations by its Jacquet module $\mathbf{r}_G^B(\pi_I)$.

Proposition 2.2. *For any $I \subset \{1, \dots, n-1\}$, the Jacquet module of π_I admits a direct sum decomposition*

$$\mathbf{r}_G^B(\pi_I) = \bigoplus_{\sigma} \sigma(\delta^{1/2}),$$

where the direct sum runs over $\sigma \in S_n$ such that $I = P_{\sigma}$ with $P_{\sigma} := \{i \in \{1, \dots, n-1\} \mid \sigma^{-1}(i) < \sigma^{-1}(i+1)\}$.

Proof. This is essentially proved in [Zel80, §2]; we omit the details. $\square$

Example 2.3. Taking $I = \{1, \dots, n-1\}$ in Proposition 2.2, we get $\mathbf{r}_G^B(\pi_I) = \mathbf{r}_G^B(\text{St}) = \delta^{1/2}$. This coincides with the usual characterization of the (generic) Steinberg representation.

By Proposition 2.2, to compute the (semi-simplified part of) cohomology of $i_1^* \mathcal{L}_{\mathbf{k}}^{\text{aut}}$, it suffices to compute that of its Jacquet module. Applying [HHS24, Corollary 2.2.1], we rewrite this Jacquet module via the constant term functor as

$$\mathbf{r}_G^B i_1^* \mathcal{L}_{\mathbf{k}}^{\text{aut}} = i_1^{*,T} \text{CT}_{B,!} \mathcal{L}_{\mathbf{k}}^{\text{aut}}.$$

But the right hand side can be computed on spectral side, where automorphic sheaf $\text{CT}_{B,!} \mathcal{L}_{\mathbf{k}}^{\text{aut}}$ corresponds to $\text{CT}_B^{\text{Spec}} \mathcal{L}_{\mathbf{k}} := \mathbf{p}_*^{\text{Spec}} \mathbf{q}^{\text{Spec},!} \mathcal{L}_{\mathbf{k}}$; the spectral behavior of $i_1^{*,T}$ is the restriction of sheaves on $\text{Par}_{\hat{T}}^{\text{unip}} = \hat{T} \times \mathbb{B}\hat{T}$ to the part with trivial $\mathbb{B}\hat{T}$.

$$\begin{array}{ccc} & \text{Par}_B^{\text{unip}} & \\ \mathbf{p}^{\text{Spec}} \swarrow & & \searrow \mathbf{q}^{\text{Spec}} \\ \hat{T} \times \mathbb{B}\hat{T} = \text{Par}_{\hat{T}}^{\text{unip}} & & \text{Par}_{\hat{G}}^{\text{unip}} \xleftarrow{\iota} V_{\hat{G},\varphi} \end{array}$$

Thus, the essential difficulty lies in computing $\mathbf{q}^{\text{Spec},!} \mathcal{L}_{\mathbf{k}}$. To compute this $!$ -pullback, take $V_{\hat{G},\varphi}^{\wedge}$ and $\text{Par}_{\hat{G}}^{\text{unip},\wedge}$ to be the formal completions of $V_{\hat{G},\varphi}$ and $\text{Par}_{\hat{G}}^{\text{unip}}$, respectively. So we have $\hat{\iota}$ and $\hat{\mathbf{q}}^{\text{Spec}}$ below.

$$\begin{array}{ccc} \text{Par}_B^{\text{unip}} & \xleftarrow{\quad} & V_{\hat{B},\varphi}^{\wedge} \\ & \searrow \hat{\mathbf{q}}^{\text{Spec}} & \searrow \\ & \text{Par}_{\hat{G}}^{\text{unip},\wedge} & \xleftarrow{\hat{\iota}} V_{\hat{G},\varphi}^{\wedge} \end{array}$$

Let $V_{\hat{B},\varphi}^{\wedge}$ be the pullback of $V_{\hat{G},\varphi}^{\wedge}$ along $\hat{\mathbf{q}}^{\text{Spec}}$. The result below describes these formal stacks.

Proposition 2.4 (Xiangqian Yang, see [Yan25, Proposition 3.12]).

(1) *As formal stacks, there is an isomorphism*

$$V_{\hat{G},\varphi}^{\wedge} \simeq (\text{Spf } \Lambda[v_1, \dots, v_{n-1}][[u_1, \dots, u_n]] / (u_1 v_1, \dots, u_{n-1} v_{n-1})) / \hat{T}.$$

- (2) For each $\sigma \in S_n$, denote $P_\sigma := \{i \in \{1, \dots, n-1\} \mid \sigma^{-1}(i) < \sigma^{-1}(i+1)\}$ and let Q_σ be its complement in $\{1, \dots, n-1\}$. Then

$$V_{\hat{B}, \varphi}^\wedge \simeq \coprod_{\sigma \in S_n} (\mathrm{Spf} \Lambda[v_i]_{i \in P_\sigma} \llbracket u_1, \dots, u_n \rrbracket / (u_i v_i)_{i \in P_\sigma}) / \hat{T}.$$

In all formal stacks above, $\hat{T}$ acts on formal variable u_i trivially with weight 0 and acts on v_i with weight $\alpha_i \in \mathbf{X}^*(\hat{T})$.

Proof. We morally sketch the proof idea in [Yan25] in the special case $G = \mathrm{GL}_2$. When $n = 2$, we have $\delta^{1/2} = \mathrm{diag}(q^{-1/2}, q^{1/2})$. Given a Weil–Deligne parameter (φ, N) where φ is semisimple, it corresponds to a point (x, y) on completed Vogan variety with condition $\mathrm{ad} \varphi(\mathrm{Fr}) \cdot N = q^{-1}N$, described by

$$V_{\hat{G}, \varphi}^\wedge = \{x = \begin{pmatrix} u_1 & 0 \\ 0 & u_2 \end{pmatrix}, y = \begin{pmatrix} 1 & v_1 \\ 0 & 1 \end{pmatrix} \mid u_1 \equiv q^{-1/2}, u_2 \equiv q^{1/2}, (qu_1 - u_2)v_1 = 0\} / \hat{T}.$$

Here $\hat{T}$ acts on x trivially, and acts on y by weight α_1 . This proves assertion (1) for GL_2 .

For assertion (2), up to $\tilde{G}$ -conjugacy, the Weyl group $W = S_2$ permutes diagonal entries of x , and hence $V_{\hat{B}, \varphi}^\wedge$ consists of two components

$$V_{\hat{B}, \varphi}^\wedge = X_\emptyset \sqcup X_{\{1\}}.$$

Here $X_\emptyset$ corresponds to $\sigma = \mathrm{id}$ and $Q_\sigma = \emptyset$, on which the points are (x, y) such that $x \equiv \mathrm{diag}(q^{-1/2}, q^{1/2})$ and $y \in B$; notice that on $X_\emptyset$, we always have $qu_1 - u_2 = 0$ so the variable v_1 is free. Also, $X_{\{1\}}$ corresponds to $\sigma = (12)$ and $Q_\sigma = \{1\}$, on which $x \equiv \mathrm{diag}(q^{1/2}, q^{-1/2})$ and $y \in B$; but $qu_1 - u_2 \neq 0$ on $X_{\{1\}}$, so the condition forces $v_1 = 0$. This completes the proof for $G = \mathrm{GL}_2$. $\square$

Notation 2.5. To further simplify the notation, write

$$X_\sigma := (\mathrm{Spf} \Lambda[v_i]_{i \in P_\sigma} \llbracket u_1, \dots, u_n \rrbracket / (u_i v_i)_{i \in P_\sigma}) / \hat{T}.$$

Then each X_σ is defined by setting $v_j = 0$ for $j \in Q_\sigma$ inside $V_{\hat{G}, \varphi}^\wedge$; in particular, $X := X_{\mathrm{id}} = V_{\hat{G}, \varphi}^\wedge$. So X_σ is viewed as a closed substack via $i_\sigma: X_\sigma \hookrightarrow X$. As for the global sections, we take

$$\mathcal{O}_{X_\sigma} = \Lambda[v_i]_{i \in P_\sigma} \llbracket u_1, \dots, u_n \rrbracket / (u_i v_i)_{i \in P_\sigma}$$

and also set $\mathcal{O}_X / \mathbf{u} := \mathcal{O}_{X_{\mathrm{id}}} / (u_1, \dots, u_n) = \Lambda[v_1, \dots, v_{n-1}]$.

Throughout the notes, we use the convention that the automorphic Vogan sheaf $\mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}$ corresponds to the spectral weight sheaf $(\mathcal{O}_X / \mathbf{u})(\mu_{\mathbf{k}})$ for $\mu_{\mathbf{k}} = k_1 e_1 + \dots + k_n e_n = m_1 \alpha_1 + \dots + m_{n-1} \alpha_{n-1}$.

Now, with notations above, the unipotent categorical local Langlands functor $\mathbb{L}^{\mathrm{unip}}$ (which is an equivalence in [Zhu25]) interpolates the equality

$$\mathbb{L}^{\mathrm{unip}}(\mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}) = \hat{i}_*(\mathcal{O}_X / \mathbf{u})(\mu_{\mathbf{k}}) \in \mathrm{Coh}(\mathrm{Par}_{\hat{G}}^{\mathrm{unip}, \wedge}).$$

On the other hand, to input the condition $b = 1$ (so that $G_b = G$), recall that the Iwahori compact induction of the trivial representation corresponds to the unipotent coherent Springer sheaf, i.e.,

$$\mathbb{L}^{\mathrm{unip}}(\mathrm{c}\text{-}\mathrm{Ind}_I^G \mathbb{1}) = \mathrm{CohSpr}^{\mathrm{unip}} := \hat{\mathfrak{q}}_*^{\mathrm{Spec}} \mathcal{O}_{\mathrm{Par}_{\hat{B}}^{\mathrm{unip}}} \in \mathrm{Coh}(\mathrm{Par}_{\hat{G}}^{\mathrm{unip}, \wedge}).$$

For our purpose of computing $i_1^* \mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}$, it suffices to compute

$$\mathrm{RHom}(\mathrm{c}\text{-}\mathrm{Ind}_I^G \mathbb{1}, \mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}) \cong \mathrm{RHom}(\mathrm{CohSpr}^{\mathrm{unip}}, \hat{i}_*(\mathcal{O}_X / \mathbf{u})(\mu_{\mathbf{k}})),$$

where the isomorphism is due to the full faithfulness of $\mathbb{L}^{\mathrm{unip}}$. However, as a consequence of Proposition 2.4(2), there is a natural isomorphism

$$\hat{i}^!(\mathrm{CohSpr}^{\mathrm{unip}}) \cong \bigoplus_{\sigma \in S_n} \mathcal{O}_{X_\sigma}$$

of ind-coherent sheaves on X . Here each $\mathcal{O}_{X_\sigma}$ is viewed as an $\mathcal{O}_X$ -module via the closed embedding $i_\sigma: X_\sigma \hookrightarrow X$.

To summarize, combining all the arguments above, the computation of $i_1^* \mathcal{L}_{\mathbf{k}}^{\mathrm{aut}}$ reduces to computing the direct sum of derived complexes

$$(\dagger) \quad \bigoplus_{\sigma \in S_n} \mathrm{RHom}(\mathcal{O}_{X_\sigma}, (\mathcal{O}_X / \mathbf{u})(\mu_{\mathbf{k}})).$$

Remark 2.6. In (†), there are $|S_n| = n!$ direct summands, and each summand is determined by $P_\sigma \subset \{1, \dots, n-1\}$ that has in total $2^{n-1} < n!$ different choices. Indeed, there is an explicit combinatorial formula for the fiber size of the map $\sigma \mapsto P_\sigma$ at each subset $I \subset \{1, \dots, n-1\}$. Moreover, this fiber size is equal to the multiplicity of π_I appearing in the cohomology $H^*(i_1^* \mathcal{L}_k^{\text{aut}})$. (Granting Theorem 2.1, we expect the cohomology of $\text{RHom}(\mathcal{O}_{X_\sigma}, (\mathcal{O}_X/\mathbf{u})(\mu_k))$ to be concentrated in certain single degree, and so also is that of $i_1^* \mathcal{L}_k^{\text{aut}}$; but these two degrees are a priori not necessarily the same.)

Suppose $I = P_\sigma$ for some $\sigma \in S_n$. On the spectral side, this multiplicity of π_I can also be interpreted as the multiplicity of the image of X_σ , as a closed substack of $\text{Par}_{\hat{B}}^{\text{unip}, \wedge}$, inside $\text{Par}_{\hat{T}}^{\text{unip}, \wedge} = \hat{T} \times \mathbb{B}\hat{T}$ along $\mathfrak{p}^{\text{Spec}}$. On the automorphic side, in the context of Yang–Zhu’s proof of torsion vanishing [YZ25], taking the global section of $\hat{T}$ -part in $\text{Par}_{\hat{T}}^{\text{unip}, \wedge}$ gives a commutative Λ -algebra $\mathcal{O}(\hat{T})$ (also denoted by $\mathcal{R}$ in [Yan25]), which essentially corresponds to the universal unramified character of T . This $\mathcal{O}(\hat{T})$ acts on $\text{CohSpr}^{\text{unip}}$ and hence on (†) above. Using this setting, we identify the multiplicity of π_I in $H^*(i_1^* \mathcal{L}_k^{\text{aut}})$ with the length of $H^*(\text{RHom}(\mathcal{O}_{X_\sigma}, (\mathcal{O}_X/\mathbf{u})(\mu_k)))$ as an $\mathcal{O}(\hat{T})$ -module.

2.2. A projective resolution. Continue with the computation of (†). We aim to compute cohomology of $\text{RHom}(\mathcal{O}_{X_\sigma}, (\mathcal{O}_X/\mathbf{u})(\mu_k))$ for each $\sigma \in S_n$. The technical strategy is to construct a free projective resolution

$$\mathcal{P}_\bullet \longrightarrow \mathcal{O}_{X_\sigma} = \mathcal{O}_X / (v_j)_{j \in Q_\sigma} \longrightarrow 0,$$

of $\mathcal{O}_X$ -modules, satisfying the $\hat{T}$ -equivariant condition.

Construction 2.7. Fix an arbitrary subset $Q \subset \{1, \dots, n-1\}$. Then $Q = Q_\sigma$ for some $\sigma \in S_n$. We use the following notations.

- Define $\mathbb{N}^Q := \{\mathbf{d} = (d_i)_{i \in Q} \mid d_i \in \mathbb{Z}_{\geq 0}\}$;
- For each $\mathbf{d} \in \mathbb{N}^Q$, write $|\mathbf{d}| := \sum_{i \in Q} d_i$ and define the weight

$$\chi_{\mathbf{d}} := \sum_{i \in Q} \lceil d_i/2 \rceil \alpha_i.$$

For $t \geq 0$, construct the t -th term of $\mathcal{P}_\bullet$ by

$$\mathcal{P}_t := \bigoplus_{\mathbf{d} \in \mathbb{N}^Q, |\mathbf{d}|=t} \mathcal{O}_X(\chi_{\mathbf{d}}).$$

This $\mathcal{P}_t$ is a free $\mathcal{O}_X$ -module with a basis denoted by $\{\mathbf{e}_{\mathbf{d}}\}_{|\mathbf{d}|=t}$. When $t = 0$, the condition $|\mathbf{d}| = 0$ forces $d_i = 0$ for all $i \in Q$, and thus $\mathcal{P}_0 = \mathcal{O}_X$. For each $\sigma \in S_n$, the projection map $\mathcal{P}_0 \rightarrow \mathcal{O}_{X_\sigma}$ is given by the natural quotient setting $v_i = 0$ for $i \in Q_\sigma$.

Next, we construct the differential map $\partial: \mathcal{P}_{t+1} \rightarrow \mathcal{P}_t$ by giving the image of each $\mathbf{e}_{\mathbf{d}}$ as follows:

$$\partial(\mathbf{e}_{\mathbf{d}}) = \sum_{i \in Q, d_i > 0} (-1)^{\varepsilon(i, \mathbf{d})} \vartheta(d_i) \cdot \mathbf{e}_{\mathbf{d} - \mathbf{e}_i}.$$

The notations in this formula are explained below.

- Given variables v_i, u_i in $\mathcal{O}_X$, we set $\vartheta_i(r) = v_i$ when r is odd, and set $\vartheta_i(r) = u_i$ when r is even.
- Define $\varepsilon(i, \mathbf{d}) := \sum_{j \in Q_{<i}} d_j$, where $Q_{<i} = Q \cap \{1, \dots, i-1\}$; we also declare $\varepsilon(1, \mathbf{d}) = 0$.
- The $\mathbf{e}_i$ is the $|Q|$ -tuple with 1 on its i -th coordinate and 0 elsewhere; set $\mathbf{e}_{\mathbf{d} - \mathbf{e}_i} = 0$ if any coordinate of $\mathbf{d} - \mathbf{e}_i$ is negative.

Note that the resolution $\mathcal{P}_\bullet$ depends on Q , but we omit Q from the notation.

Lemma 2.8. *The differential map $\partial: \mathcal{P}_{t+1} \rightarrow \mathcal{P}_t$ in Construction 2.7 satisfies $\partial \circ \partial = 0$.*

Proof. We compute $(\partial \circ \partial)(\mathbf{e}_{\mathbf{d}})$. This is a linear combination of elements $\mathbf{e}_{\mathbf{d} - \mathbf{e}_i - \mathbf{e}_j}$. When $i = j$, the coefficient of $\mathbf{e}_{\mathbf{d} - 2\mathbf{e}_i}$ is given by

$$(-1)^{\varepsilon(i, \mathbf{d}) + \varepsilon(i, \mathbf{d} - \mathbf{e}_i)} \vartheta_i(d_i) \vartheta_i(d_i - 1) = (-1)^{\varepsilon(i, \mathbf{d}) + \varepsilon(i, \mathbf{d} - \mathbf{e}_i)} u_i v_i = 0,$$

because we have $u_i v_i = 0$ in $\mathcal{O}_X$. So it only remains to consider the case $i \neq j$. Note that there are two ways to get $\mathbf{e}_{\mathbf{d} - \mathbf{e}_i - \mathbf{e}_j}$ from $\mathbf{e}_{\mathbf{d}}$, namely through either $\mathbf{e}_{\mathbf{d} - \mathbf{e}_i}$ or $\mathbf{e}_{\mathbf{d} - \mathbf{e}_j}$. So the coefficient of $\mathbf{e}_{\mathbf{d} - \mathbf{e}_i - \mathbf{e}_j}$

in $(\partial \circ \partial)(\mathbf{e}_{\mathbf{d}})$ is computed as

$$(-1)^{\varepsilon(i, \mathbf{d}) + \varepsilon(j, \mathbf{d} - \mathbf{e}_i)} \vartheta_i(d_i) \vartheta_j(d_j) + (-1)^{\varepsilon(j, \mathbf{d}) + \varepsilon(i, \mathbf{d} - \mathbf{e}_j)} \vartheta_j(d_j) \vartheta_i(d_i).$$

But we always have $\varepsilon(j, \mathbf{d} - \mathbf{e}_i) = \varepsilon(j, \mathbf{d}) - \mathbb{1}_{i < j}$, where $\mathbb{1}_{i < j} \in \{0, 1\}$ is the characteristic function to test whether $i < j$ is true or not; similarly, $\varepsilon(i, \mathbf{d} - \mathbf{e}_j) = \varepsilon(i, \mathbf{d}) - \mathbb{1}_{j < i}$. Thus, for arbitrary i and j , the two exponents of -1 above always differ by 1, so this coefficient equals zero. $\square$

Lemma 2.9. *The resolution $\mathcal{P}_\bullet$ in Construction 2.7 is $\hat{T}$ -equivariant.*

Proof. Recall that $\hat{T}$ acts on v_i with weight α_i and acts on u_i with weight 0. Fix $\chi \in \mathbf{X}^*(\hat{T})$. We claim that the following two multiplication maps of $\mathcal{O}_X$ -modules are $\hat{T}$ -equivariant:

$$v_i: \mathcal{O}_X(\chi) \longrightarrow \mathcal{O}_X(\chi - \alpha_i), \quad u_i: \mathcal{O}_X(\chi) \longrightarrow \mathcal{O}_X(\chi).$$

Indeed, it suffices to consider the case $\chi = m\alpha_i$ for some $m \in \mathbb{Z}_{\geq 0}$, so $\chi - \alpha_i = (m-1)\alpha_i$. To check the first map, for $t \in \hat{T}$, note that $t.(v_i a) = \alpha_i(t) \cdot v_i(t.a)$ for any section a in $\mathcal{O}_X$. So in $\mathcal{O}_X((m-1)\alpha_i)$, we have $t.(v_i a) = \alpha_i^{m-1}(t) \cdot t.(v_i a) = \alpha_i^m(t) \cdot v_i(t.a)$. On the other hand, inside $\mathcal{O}_X(m\alpha_i)$, we have $v_1(t.a) = v_1(\alpha_i^m(t) \cdot (t.a)) = \alpha_i^m(t) \cdot v_i(t.a)$. Comparing these two right hand sides, we identify $t.(v_i a)$ in $\mathcal{O}_X((m-1)\alpha_i)$ and $v_i(t.a)$ in $\mathcal{O}_X(m\alpha_i)$. This gives the $\hat{T}$ -equivariant property of the multiplication by v_i , and that by u_i can be checked similarly. So we have verified the claim.

Back to our construction of $\mathcal{P}_\bullet$. When d_i is odd (resp. even), we have $\vartheta_i(d_i) = v_i$ of $\hat{T}$ -weight α_i (resp. $\vartheta_i(d_i) = u_i$ of $\hat{T}$ -weight 0), and thus $v_i: \mathcal{O}_X(\chi_{\mathbf{d}}) \rightarrow \mathcal{O}_X(\chi_{\mathbf{d}} - \alpha_i)$ (resp. $u_i: \mathcal{O}_X(\chi_{\mathbf{d}}) \rightarrow \mathcal{O}_X(\chi_{\mathbf{d}})$) is $\hat{T}$ -equivariant by the claim. But when d_i is odd (resp. even), there must be $\lceil (d_i - 1)/2 \rceil = \lceil d_i/2 \rceil - 1$ (resp. $\lceil (d_i - 1)/2 \rceil = \lceil d_i/2 \rceil$), so the target of multiplication by v_i (resp. multiplication by u_i) equals $\mathcal{O}_X(\chi_{\mathbf{d}} - \alpha_i) = \mathcal{O}_X(\chi_{\mathbf{d} - \mathbf{e}_i})$ (resp. $\mathcal{O}_X(\chi_{\mathbf{d}}) = \mathcal{O}_X(\chi_{\mathbf{d} - \mathbf{e}_i})$), which is the same as in $\mathcal{P}_\bullet$. $\square$

Example 2.10. We illustrate the resolution in Construction 2.7 for GL_2 and GL_3 .

(1) For $G = GL_2$, the Weyl group $W = S_2 = \{\text{id}, (12)\}$; we have $Q_{\text{id}} = \emptyset$ and $Q_{(12)} = \{1\}$. Only the latter case is nontrivial so we take $Q = Q_{(12)}$. Then

$$\mathbf{d} = d_1 \in \mathbb{Z}_{\geq 0} = \mathbb{N}^Q, \quad \chi_{\mathbf{d}} = \lceil d_1/2 \rceil \alpha_1.$$

So the resolution is 2-periodic, given by

$$\mathcal{P}_t = \mathcal{O}_X(\lceil t/2 \rceil \alpha_1).$$

As for the differential map, we compute $\varepsilon(i, \mathbf{d}) = \varepsilon(i, d_1) = 0$, and hence

$$\partial(\mathbf{e}_{\mathbf{d}}) = \partial(\mathbf{e}_{d_1}) = \vartheta_1(d_1) \cdot \mathbf{e}_{d_1-1}.$$

Recall that $\vartheta_1(d_1) = v_1$ for $2 \nmid d_1$ and u_1 for $2 \mid d_1$. To conclude, the resolution is

$$\begin{array}{ccccccc} \cdots & \xrightarrow{u_1} & \mathcal{O}_X(2\alpha_1) & \xrightarrow{v_1} & \mathcal{O}_X(\alpha_1) & \xrightarrow{u_1} & \mathcal{O}_X(\alpha_1) \xrightarrow{v_1} \mathcal{O}_X \twoheadrightarrow \mathcal{O}_{X(12)} \longrightarrow 0. \\ & & \parallel & & \parallel & & \parallel \\ & & \mathcal{P}_3 & & \mathcal{P}_2 & & \mathcal{P}_1 & & \mathcal{P}_0 \end{array}$$

As a remark, this recovers the resolution constructed in [BM23, §2] for the PGL_2 case (which is essentially the same as the GL_2 case).

(2) For $G = GL_3$, the case of $Q = \{1\}$ or $\{2\}$ is essentially the same as in (1), and that of $Q = \emptyset$ is trivial. So we only need to consider $Q = \{1, 2\}$. In this case,

$$\mathbf{d} = (d_1, d_2) \in \mathbb{Z}_{\geq 0}^2 = \mathbb{N}^Q, \quad \chi_{\mathbf{d}} = \lceil d_1/2 \rceil \alpha_1 + \lceil d_2/2 \rceil \alpha_2.$$

Then we have

$$\mathcal{P}_t = \bigoplus_{d_1 + d_2 = t} \mathcal{O}_X(\lceil d_1/2 \rceil \alpha_1 + \lceil d_2/2 \rceil \alpha_2),$$

where $\mathcal{P}_t$ consists of exactly $t + 1$ direct summands, and each of which has a basis $\mathbf{e}_{\mathbf{d}} = \mathbf{e}_{(d_1, d_2)}$.

To get the formula of $\partial(\mathbf{e}_{\mathbf{d}})$, first compute $\varepsilon(1, \mathbf{d}) = 0$ and $\varepsilon(2, \mathbf{d}) = d_1$; it then follows that

$$\partial(\mathbf{e}_{(d_1, d_2)}) = \vartheta_1(d_1) \mathbf{e}_{(d_1-1, d_2)} + (-1)^{d_1} \vartheta_2(d_2) \mathbf{e}_{(d_1, d_2-1)}.$$

In particular, the map $\partial: \mathcal{P}_{t+1} \rightarrow \mathcal{P}_t$ can be illustrated by a matrix of size $(t+1) \times (t+2)$ with all elements lying in $\{\pm v_i, \pm u_i, 0\}$. This map will be significantly simplified if we work in $\mathcal{O}_X/\mathbf{u}$ with setting all $u_i = 0$.

2.3. Proof of Theorem 2.1. We proceed to compute the cohomology

$$H^*(\mathrm{RHom}(\mathcal{O}_{X_\sigma}, (\mathcal{O}_X/\mathbf{u})(\mu_{\mathbf{k}})))$$

of each summand of (†) using the resolution in Construction 2.7. In $\mathrm{IndCoh}(\mathrm{Par}_{\hat{G}}^{\mathrm{unip}, \wedge})$, applying the internal Hom functor to

$$\mathcal{P}_\bullet \longrightarrow \mathcal{O}_{X_\sigma} \longrightarrow 0,$$

and then taking the $\hat{T}$ -invariant part, we get a cochain complex

$$(\mathcal{C}^\bullet)^{\hat{T}} := \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{P}_\bullet, (\mathcal{O}_X/\mathbf{u})(\mu_{\mathbf{k}}))^{\hat{T}},$$

whose t -th term reads

$$\begin{aligned} (\mathcal{C}^\bullet)^{\hat{T}} &\simeq \bigoplus_{\mathbf{d} \in \mathbb{N}^Q, |\mathbf{d}|=t} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X(\chi_{\mathbf{d}}), (\mathcal{O}_X/\mathbf{u})(\mu_{\mathbf{k}}))^{\hat{T}} \\ &\cong \bigoplus_{\mathbf{d} \in \mathbb{N}^Q, |\mathbf{d}|=t} (\mathcal{O}_X/\mathbf{u})(\mu_{\mathbf{k}} - \chi_{\mathbf{d}})^{\hat{T}}. \end{aligned}$$

Here the second equality uses the formula of $\mathcal{P}_\bullet$. Note that the differential map in $(\mathcal{C}^\bullet)^{\hat{T}}$ is directly induced from ∂ in $\mathcal{P}_\bullet$.

To further compute $(\mathcal{O}_X/\mathbf{u})(\mu_{\mathbf{k}} - \chi_{\mathbf{d}})^{\hat{T}}$, we need the following result.

Lemma 2.11. *Let $\nu = r_1\alpha_1 + \cdots + r_{n-1}\alpha_{n-1}$ for $r_1, \dots, r_{n-1} \in \mathbb{Z}$. Then we have*

$$\mathcal{O}_X(\nu)^{\hat{T}} = \begin{cases} 0, & \text{if } r_j > 0 \text{ for some } j; \\ v_1^{-r_1} \cdots v_{n-1}^{-r_{n-1}} \Lambda[u_n][u_j]_{r_j=0}, & \text{otherwise.} \end{cases}$$

In particular, $(\mathcal{O}_X/\mathbf{u})(\nu)^{\hat{T}}$ is either isomorphic to Λ or 0, determined by whether all $r_i \leq 0$ or not.

Proof. Since the $\hat{T}$ -weights of v_i and u_i are respectively α_i and 0, for any monomial of the form $v_1^{a_1} \cdots v_{n-1}^{a_{n-1}} \cdot u_1^{b_1} \cdots u_n^{b_n}$ in $\mathcal{O}_X$, after twisting by $\mathcal{O}_X(\nu)$, its $\hat{T}$ -weight equals $\sum_{1 \leq j \leq n-1} (r_j + a_j)\alpha_j$. So the $\hat{T}$ -invariant condition forces $r_j + a_j = 0$, or equivalently $a_j = -r_j \geq 0$. Also, we have conditions $v_j u_j = 0$ in $\mathcal{O}_X$, so we see if $r_j < 0$ (and hence $a_j > 0$), then no u_j is allowed, meaning that $b_j = 0$ for all j ; else if $r_j = 0$ (and hence $a_j = 0$), the u_j is free. This completes the proof. $\square$

Corollary 2.12. *Fix $Q \subset \{1, \dots, n-1\}$ as before. If $(\mathcal{C}^\bullet)^{\hat{T}} \neq 0$, then there exists $\mathbf{d} \in \mathbb{N}^Q$ such that for all $i \in Q$, we have $\lceil d_i/2 \rceil \geq m_i = k_1 + \cdots + k_i$.*

Proof. This follows immediately from the condition $\mu_{\mathbf{k}} - \chi_{\mathbf{d}} \leq 0$ given by Lemma 2.11. $\square$

For the proof of Theorem 2.1, we point out that the case of GL_3 already exhibits all phenomena present for GL_n . Thus, in the following, we only consider $n = 3$. There are $2^{n-1} = 4$ different choices of Q_σ , and we compute the cohomology or the desired $\mathrm{RHom}(\mathcal{O}_{X_\sigma}, (\mathcal{O}_X/\mathbf{u})(\mu_{\mathbf{k}}))$ case by case.

Case I. Let $Q_\sigma = \emptyset$. This uniquely corresponds to $\sigma = \mathrm{id}$, so $X_\sigma = X$. Applying Lemma 2.11 (together with the formula of $(\mathcal{C}^\bullet)^{\hat{T}}$ before), we directly get

$$\mathrm{RHom}(\mathcal{O}_X, (\mathcal{O}_X/\mathbf{u})(\mu_{\mathbf{k}})) \cong \begin{cases} 0, & \text{if } m_1 > 0 \text{ or } m_2 > 0, \\ \Lambda, & \text{if } m_1 \leq 0 \text{ and } m_2 \leq 0. \end{cases}$$

Therefore, for $\mathbf{k}$ of degree 0 such that $m_1, m_2 \leq 0$, we have $I_{\mathbf{k}} = \{1, 2\}$ and $J_{\mathbf{k}} = \emptyset$ by definition. It gives rise to the generic Steinberg representation $\pi_{\{1,2\}} = \mathrm{St}$ without cohomological degree shift.

Case II. Let $Q_\sigma = \{1\}$, corresponding to $\sigma \in \{(12), (123)\} \subset S_3$. In this case, $\mathcal{O}_{X_\sigma} = \mathcal{O}_X/v_1$. The resolution of $\mathcal{O}_{X_\sigma}$ is essentially the same as Example 2.10(1) for the GL_2 case, namely $\mathcal{P}_\bullet \rightarrow \mathcal{O}_{X_\sigma} \rightarrow 0$ with $\mathcal{P}_t = \mathcal{O}_X(\lceil t/2 \rceil \alpha_1)$. Applying Lemma 2.11, we compute

$$\begin{aligned} (\mathcal{C}^i)^{\hat{T}} &:= \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{P}_i, (\mathcal{O}_X/\mathbf{u})(\mu_{\mathbf{k}}))^{\hat{T}} \\ &= (\mathcal{O}_X/\mathbf{u})((m_1 - \lceil i/2 \rceil)\alpha_1 + m_2\alpha_2)^{\hat{T}} \\ &\cong \begin{cases} 0, & \text{if } m_1 - \lceil i/2 \rceil > 0 \text{ or } m_2 > 0, \\ \Lambda, & \text{if } m_1 - \lceil i/2 \rceil \leq 0 \text{ and } m_2 \leq 0. \end{cases} \end{aligned}$$

In particular, if $(\mathcal{C}^i)^{\hat{T}} \neq 0$, then $m_1 \leq \lceil i/2 \rceil$; this condition is equivalent to $i \geq 2m_1 - 1$ (and we always have $i \geq 0$). So the whole complex, which is bounded below, is written as

$$(\mathcal{C}^\bullet)^{\hat{T}} = \left[0 \rightarrow \cdots \rightarrow 0 \rightarrow \Lambda \xrightarrow{u_1} \Lambda \xrightarrow[\sim]{v_1} \Lambda \xrightarrow{u_1} \Lambda \xrightarrow[\sim]{v_1} \cdots \right].$$

Here the first term sits in degree 0 and the first nonzero term sits in degree $2m_1 - 1$. This complex is bounded below. Each multiplication by v_1 is an isomorphism, whereas each multiplication by u_1 is zero as the base ring is $\mathcal{O}_X/\mathbf{u}$. Therefore, taking cohomology on the complex $(\mathcal{C}^\bullet)^{\hat{T}}$, we see exactly two possibilities:

- (2a) If $m_1 \leq 0$, then the cohomology of $(\mathcal{C}^\bullet)^{\hat{T}}$ vanishes at all degrees.
- (2b) If $m_1 > 0$ and $m_2 \leq 0$, the cohomology is non-vanishing only at the first nonzero term, i.e., $H^*((\mathcal{C}^\bullet)^{\hat{T}}) = 0$ except for degree $2m_1 - 1$, where

$$H^{2m_1-1}((\mathcal{C}^\bullet)^{\hat{T}}) = \Lambda.$$

To conclude, we explain the meaning of this cohomology. For $\mathbf{k}$ of degree 0 such that $m_1 > 0$ and $m_2 \leq 0$, we have $I_{\mathbf{k}} = \{2\}$ and $J_{\mathbf{k}} = \{1\}$ by definition. So this cohomology gives the representation $\pi_{\{2\}}[1 - 2m_1]$, as predicted by Conjecture 1.4.

Case III. Let $Q_\sigma = \{2\}$, corresponding to $\sigma \in \{(23), (132)\} \subset S_3$. This is the same computation as in Case II, with only swapping the two indices m_1, m_2 and v_1, v_2 . Thus, we also get exactly two possibilities of the cohomology of $(\mathcal{C}^\bullet)^{\hat{T}}$, namely:

- (3a) If $m_2 \leq 0$, then $H^*((\mathcal{C}^\bullet)^{\hat{T}}) = 0$ at all degrees.
- (3b) If $m_1 \leq 0$ and $m_2 > 0$, then $H^*((\mathcal{C}^\bullet)^{\hat{T}}) = 0$ except for degree $2m_2 - 1$, where

$$H^{2m_2-1}((\mathcal{C}^\bullet)^{\hat{T}}) = \Lambda.$$

Again, for $\mathbf{k}$ of degree 0 such that $m_1 \leq 0$ and $m_2 > 0$, this corresponds to $I_{\mathbf{k}} = \{1\}$, $J_{\mathbf{k}} = \{2\}$, and hence gives the representation $\pi_{\{1\}}[1 - 2m_2]$ as predicted by Conjecture 1.4.

Case IV (The difficult case). Let $Q_\sigma = \{1, 2\}$, corresponding to $\sigma \in \{(23), (132)\} \subset S_3$. Following Construction 2.7 and Example 2.10(2), we write down the resolution $\mathcal{P}_\bullet$ of $\mathcal{O}_{X_\sigma} = \mathcal{O}_X/(v_1, v_2)$, that is,

$$\mathcal{P}_t = \bigoplus_{d_1+d_2=t} \mathcal{O}_X(\lceil d_1/2 \rceil \alpha_1 + \lceil d_2/2 \rceil \alpha_2).$$

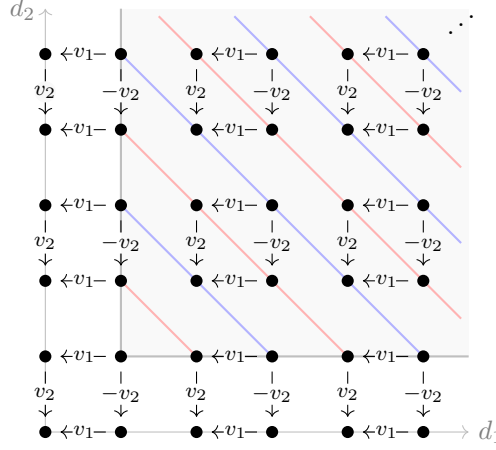
Using Lemma 2.11, whenever $(\mathcal{C}^i)^{\hat{T}} := \text{Hom}_{\mathcal{O}_X}(\mathcal{P}_i, (\mathcal{O}_X/\mathbf{u})(\mu_{\mathbf{k}}))^{\hat{T}} \neq 0$, we must have $\lceil d_1/2 \rceil \geq m_1$ and $\lceil d_2/2 \rceil \geq m_2$. Thus, the minimal possible values of d_i for $i \in \{1, 2\}$ are given by

$$d_{i,\min} = \begin{cases} 0, & m_i \leq 0, \\ 2m_i - 1, & m_i \geq 1. \end{cases}$$

We then describe the differential map. Since we eventually work over $\mathcal{O}_X/\mathbf{u}$, to simplify the argument, we take all $u_i = 0$ in the formula of $\partial(\mathbf{e}_{\mathbf{d}})$ in Construction 2.7. Then the map ∂ over $\mathcal{O}_X/\mathbf{u}$ is read as

$$\mathbf{e}_{(d_1, d_2)} \mapsto \begin{cases} 0, & \text{if } 2 \mid d_1 \text{ and } 2 \mid d_2, \\ v_2 \cdot \mathbf{e}_{(d_1, d_2-1)}, & \text{if } 2 \mid d_1 \text{ and } 2 \nmid d_2, \\ v_1 \cdot \mathbf{e}_{(d_1-1, d_2)}, & \text{if } 2 \nmid d_1 \text{ and } 2 \mid d_2, \\ v_1 \cdot \mathbf{e}_{(d_1-1, d_2)} - v_2 \cdot \mathbf{e}_{(d_1, d_2-1)}, & \text{if } 2 \nmid d_1 \text{ and } 2 \nmid d_2. \end{cases}$$

The following picture depicts the differential map above. The node at coordinate (d_1, d_2) represents the direct summand $\mathcal{O}_X(\lceil d_1/2 \rceil \alpha_1 + \lceil d_2/2 \rceil \alpha_2)$ in $\mathcal{P}_{d_1+d_2}$. Each arrow is given by a multiplication. If there are no arrows between two nodes, then the differential map is zero. When $m_1, m_2 \geq 1$, the gray-shaded part in the picture indicates the part of $d_i \geq d_{i,\min}$ for $i \in \{1, 2\}$; note that in this case $2 \nmid d_{i,\min}$.



Therefore, taking cohomology on this bicomplex, we see exactly two possibilities again:

- (4a) If $m_1 \leq 0$ or $m_2 \leq 0$, we have $d_{1,\min} = 0$ or $d_{2,\min} = 0$, and then $H^*((\mathcal{C}^i)^{\hat{T}}) = 0$ at all degrees.
- (4b) If $m_1, m_2 > 0$, the cohomology is non-vanishing only at the corner $(d_{1,\min}, d_{2,\min})$. It follows that $H^*((\mathcal{C}^i)^{\hat{T}}) = 0$ except at the degree $d_{1,\min} + d_{2,\min} = (2m_1 - 1) + (2m_2 - 1)$, where

$$H^{(2m_1-1)+(2m_2-1)}((\mathcal{C}^i)^{\hat{T}}) = \Lambda.$$

Similar to the explanation given in the three cases before, for $\mathbf{k}$ of degree 0 such that $m_1, m_2 > 0$, we have $I_{\mathbf{k}} = \emptyset$ and $J_{\mathbf{k}} = \{1, 2\}$. As $\pi_{\emptyset} = \mathbf{1}$, the trivial representation of $G(\mathbb{Q}_p)$, this cohomology corresponds to the representation $\mathbf{1}[(1 - 2m_1) + (1 - 2m_2)]$, as desired.

This finishes the computation of cohomology of the derived complex $(\dagger)$ and proves Theorem 2.1 for $G = GL_3$. The argument for $G = GL_n$ is essentially the same.

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